

Second part: Open book

Total score: 10 pts.

You are warmly invited to be tidy and concise. Messy papers will be penalized.

6. A telecommunication company has to install some antennas to cover a region. The region is divided into a set of zones Z . The company can install antennas in a set of candidate sites S . Given a candidate site $i \in S$ and a zone $j \in Z$ it is known the level of the received signal p_{ij} if an antenna is installed in i . A zone can be served by at most one active antenna and the signal received from that antenna must be greater or equal than Δ .

Formulate the problem of maximizing the number of served zones as a linear optimization problem.

Variante: In order to avoid poor quality solutions, the company has to introduce constraints on the interference. If a zone j is served by one antenna i , the total of the received signals (minus that of the antenna serving the zone) must be less than or equal to δ .

7. Consider the diet problem below, where 1 and 2 are the indices of the possible feeds and A, B and C are the nutritional components.

$$\min \quad 4x_1 + 3x_2 \quad (1)$$

$$A) \quad 2x_1 + x_2 \geq 8 \quad (2)$$

$$B) \quad 2x_1 + 3x_2 \geq 12 \quad (3)$$

$$C) \quad x_2 \geq 1 \quad (4)$$

$$x_1, x_2 \geq 0 \quad (5)$$

Write the dual and the complementary slackness equations. Derive the complementary dual solution associated with $\bar{x} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$. Is it optimal? Justify the answer.

Let us suppose that in the market are introduced two new feeds (3 and 4) whose contributions to the nutritive components and costs are given by the following table.

Tabella 1: default

feed	3	4
A	0	2
B	4	0
C	2	2
cost	2	2

What is the feed that is worth considering in the diet in order to reduce the overall cost? Justify the answer.